SG No. 49 D_{2h}^{3} $Pccm$ [orthorhombic]

* plus set: + [0, 0, 0]

Table 1: Wyckoff site: 2a, site symmetry: ..2/m

No.	position	mapping
1	[0, 0, 0]	[1,2,5,6]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[3,4,7,8]

Table 2: Wyckoff site: 2b, site symmetry: ..2/m

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[1,2,5,6]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[3,4,7,8]

Table 3: Wyckoff site: 2c, site symmetry: ..2/m

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	[1,2,5,6]
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[3,4,7,8]

Table 4: Wyckoff site: 2d, site symmetry: ..2/m

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	[1,2,5,6]
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[3,4,7,8]

Table 5: Wyckoff site: 2e, site symmetry: 222

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	[1,2,3,4]
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	[5,6,7,8]

Table 6: Wyckoff site: 2f, site symmetry: 222

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[1,2,3,4]
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[5,6,7,8]

Table 7: Wyckoff site: 2g, site symmetry: 222

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	[1,2,3,4]

continued ...

Table 7

No.	position	mapping
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,7,8]`

Table 8: Wyckoff site: `2h`, site symmetry: `222`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,7,8]`

Table 9: Wyckoff site: `4i`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,4]`
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[2,3]`
3	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[5,8]`
4	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[6,7]`

Table 10: Wyckoff site: `4j`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4]`
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,3]`
3	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,8]`
4	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[6,7]`

Table 11: Wyckoff site: `4k`, site symmetry: `.2.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,3]`
2	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[2,4]`
3	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[5,7]`
4	$\left[0,\ y,\ \frac{3}{4}\right]$	`[6,8]`

Table 12: Wyckoff site: `4l`, site symmetry: `.2.`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ \frac{1}{4}\right]$	`[1,3]`
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{4}\right]$	`[2,4]`
3	$\left[\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	`[5,7]`
4	$\left[\frac{1}{2},\ y,\ \frac{3}{4}\right]$	`[6,8]`

Table 13: Wyckoff site: `4m`, site symmetry: `..2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[3,4]`
3	$[0,\ 0,\ -z]$	`[5,6]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[7,8]`

Table 14: Wyckoff site: `4n`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,4]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`

Table 15: Wyckoff site: `4o`, site symmetry: `..2`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,4]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[5,6]`
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`

Table 16: Wyckoff site: `4p`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,2]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[3,4]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[7,8]`

Table 17: Wyckoff site: `4q`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,6]`
2	$[-x,\ -y,\ 0]$	`[2,5]`
3	$\left[-x,\ y,\ \frac{1}{2}\right]$	`[3,8]`
4	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[4,7]`

Table 18: Wyckoff site: `8r`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`

continued ...

Table 18

No.	position	mapping
2	$[-x, -y, z]$	[2]
3	$\left[-x, y, \frac{1}{2}-z\right]$	[3]
4	$\left[x, -y, \frac{1}{2}-z\right]$	[4]
5	$[-x, -y, -z]$	[5]
6	$[x, y, -z]$	[6]
7	$\left[x, -y, z+\frac{1}{2}\right]$	[7]
8	$\left[-x, y, z+\frac{1}{2}\right]$	[8]